

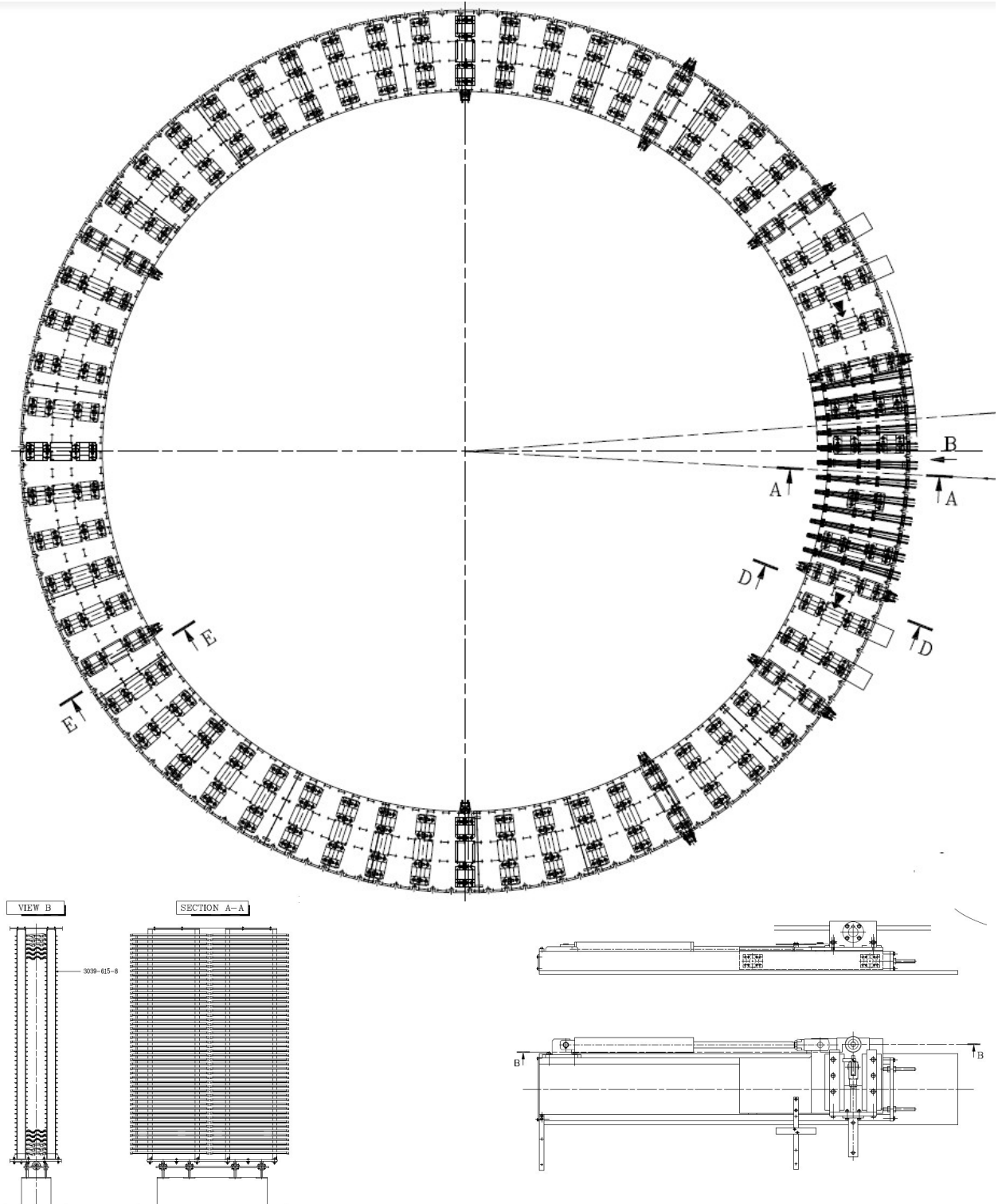
## ROTARY-RACK 45

MACHINE TYPE:HT-103 SERIAL NUMBER :M.02224

FP McCANN (Cadeby)

ORDER : C22004

YEAR OF PRODUCTION 2023



SOP M.02224.00 Contents List ROTARY-RACK 45

SOP M.02224.01 Rotary-rack 45

SOP M.02224.02 Rotary Step Indexer

## ROTARY-RACK 45

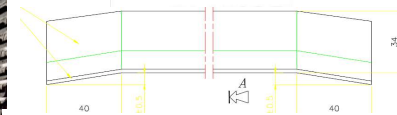
MACHINE TYPE:HT-103 SERIAL NUMBER :M.02224

## Rotary-rack 45



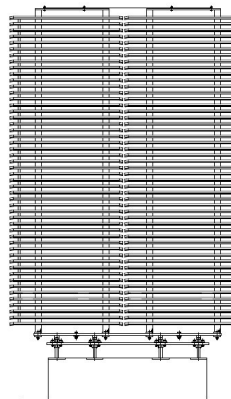
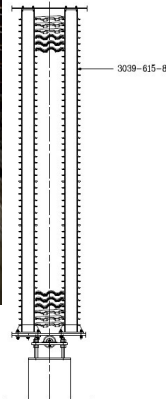
288944

## GUIDE FOR RACK



VIEW B

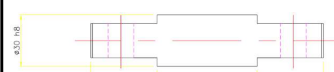
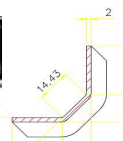
SECTION A-A



SECTION F-F

288944

## GUIDE FOR RACK

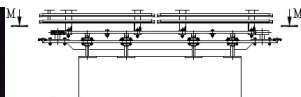


5050-059 U146

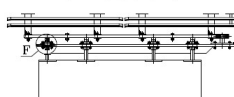
46120

## WHEEL HOLDER SHAFT

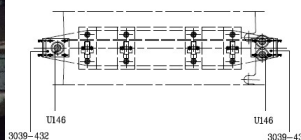
SECTION D-D



SECTION E-E



SECTION M-M



## General Description and Function

\*The Rotary Rack consists of a rotating circular structure, mounted on a fixed base with support wheels, rising across 50 independent levels or trays. Each level is equipped with paired angular guides (left/right) that hold the molds with fresh tiles. This configuration enables the simultaneous storage, transport, and curing of the entire load ensuring a uniform thermal process

## Critical Points and Risk Management

## Travel System Control:

The lower and side support wheels, along with their axles, are critical for stable rotation and overall structure alignment. Any wear or misalignment will cause vibrations, accelerated guide wear, and potential tile damage.

## Guide Integrity and Cleanliness:

\*The condition of the 30,000 angular guides is vital. Accumulated dirt (concrete residue, dust) or their wear can:

- \*Cause abrasion and damage to the fresh tile side surfaces.
- \*Alter the mould's path, causing jams or misalignment.
- \*Generate more residue through friction, creating a contamination cycle.

## Bearing Lubrication:

\*Proper greasing of the bearings in all support wheels is essential for smooth movement and to prevent premature failure due to friction.

## SUPERVISION LEVEL

## DAILY

288944 Guide for rack (Deracker / Racker)  
46120 Wheel holder shaft (Deracker / Racker)  
21647 Wheel + Bearing (it. 19500) (Deracker / Racker)  
Visual inspection of accessible guides and wheels visible from the lower pit.(Adequate lighting is required)

## WEEKLY

288944 Guide for rack (30000 units)  
46120 Wheel holder shaft (238 units)  
21647 Wheel + Bearing (it. 19500) (260 units)  
55435 Wheel support (10 units)  
55437 Wheel support (6 units)

Full perimeter inspection, verification of wheels and guides, and lubrication.

## Lubrication:

\*Apply grease to the bearings of all wheels according to manufacturer specifications.

\*Check levels in automatic systems.

## CLEANING

\*Is vital to prevent tile damage. The frequency (daily, weekly, monthly) must be determined based on observed issues (amount of concrete residue).

## For cleaning guides and structural components:

\*Loose Residue/Dust:  
\*Use a low-pressure air gun (< 3 bar) followed by dry cloths.

## Adhered Concrete Residue:

\*Use plastic or wooden scrapers and stiff-bristle brushes (never metallic) to avoid damaging surfaces.

## Critical Areas:

\*Pay special attention to the base and inner sides of the angular guides.

|                                                                                                                                  |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>VORTEX HYDRA</b>                                                                                                              | <b>STANDARD OPERATING PROCEDURE</b> | Code: SOP M.02224.02<br>Date: 15/12/25<br>Page: 3 de 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>ROTARY-RACK 45</b>                                                                                                            |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>MACHINE TYPE:HT-103 SERIAL NUMBER :M.02224</b>                                                                                |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Rotary Step Indexer</b>                                                                                                       |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <div data-bbox="150 347 507 1111"> </div> <div data-bbox="539 501 871 909"> </div>                                               |                                     | <div data-bbox="1002 322 1334 344"> <b>General Description and Function</b> </div> <p>*This assembly is the mechanism that performs the step-by-step (indexing) advance of the Rotary structure , precisely positioning it at the designated points for loading and unloading all tile moulds stored on its angles /pallet guides.</p> <div data-bbox="986 501 1350 524"> <b>Critical Points and Risk Management</b> </div> <p><b>Positioning Accuracy:</b><br/>         * The stop position must be perfect to allow pushers to insert and extract moulds without interference at the Rotary's entry and exit points.</p> <p><b>Hydraulic Operational Parameters:</b><br/>         *It is crucial to maintain the specified working pressure and correct oil level in the hydraulic power unit's reservoir. Deviations directly affect the force, speed, and accuracy of movement.</p> <div data-bbox="1078 792 1254 815"> <b>SHEET PROCESS</b> </div> <p>Process documentation must include and keep the following reference values updated:</p> <p>*Advance and stop positions of the hydraulic cylinders to confirm a consistent advance every cycle.</p> <p>* Working pressure of the hydraulic power unit.</p> <p>*Oil level to be maintained in the reservoir.</p> <p>*Advance and retraction times for each pusher, for every stop point inside the Rotary-Rack 45.</p>                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <div data-bbox="150 1155 880 1473"> </div> <div data-bbox="172 1482 858 1751"> </div> <div data-bbox="140 1832 858 2078"> </div> |                                     | <div data-bbox="1062 1106 1270 1128"> <b>SUPERVISION LEVEL</b> </div> <div data-bbox="1123 1128 1206 1151"> <b>WEEKLY</b> </div> <div data-bbox="896 1151 1430 1308"> <p>277457 Bearing CUS-LIBE CPS0023 (4 units)<br/>         18000 Bearing 6206 2RSR (2 units)<br/>         10151 Small wheel (1 u)<br/>         43653 Guide (2 units)<br/>         48479 Hydraulic cylinder with flange (1 u)<br/>         55096 Hydraulic cylinder with female hinge (1 u)<br/> <u>Check bearings, wheels, guides, and perform visual</u></p> </div> <div data-bbox="1114 1330 1216 1352"> <b>MONTHLY</b> </div> <div data-bbox="896 1352 1430 1487"> <p>276856 Fork for pneumatic cylinder (1 u)<br/>         2769 40 Fork for pneumatic cylinder (1 u)<br/>         55090 Pivot (1 u)<br/>         288962 Hydraulic power unit (ref. scheme SHO.1130.1) (1)<br/> <u>Verify forks, pivots, fasteners, connectors, and and solenoid valves.</u></p> </div> <p><b>Parameter Verification:</b><br/>         *Verify that the working pressure and oil level are within the ranges specified in the documentation</p> <p><b>Mechanical Inspection:</b><br/>         *Verify the tightness and condition of cylinder fasteners and forks.</p> <p><b>Hydraulic Inspection:</b><br/>         *Check the integrity of connectors, hydraulic hoses and solenoid valves on the power unit.</p> <p><b>Area Inspection:</b><br/>         *Look for oil contamination</p> <div data-bbox="1114 1845 1216 1890"> <b>CLEANING WEEKLY</b> </div> <p>*Surfaces and sides of the hydraulic power unit, contours of the cylinders, and their brackets.</p> <p>*Use appropriate tools (brushes, dry cloths) to remove dust, debris, and any accumulated material.</p> <p>*Prevent dirt from clogging breathers or entering connectors.</p> |